Vector Databases and LLM

Advanced Natural Language Processing
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Hugging Face

- **The Model Hub:** A central repository where we can access all available models and delve into their features.
- **The Transformers Libraries:** These powerful libraries enable us to effortlessly leverage the models housed within the Hub.

Hugging Face

- **Tasks:** The tasks for which the model has been trained.
- **Libraries:** Libraries that the models are compatible with: almost all of them are compatible with PyTorch, and a big part of the major ones also support TensorFlow.
- **Datasets:** Datasets that have been used to train the model.
- **Languages:** Languages the model has been trained with.
- **Licenses:** The license under which the model is distributed.

Kaggle

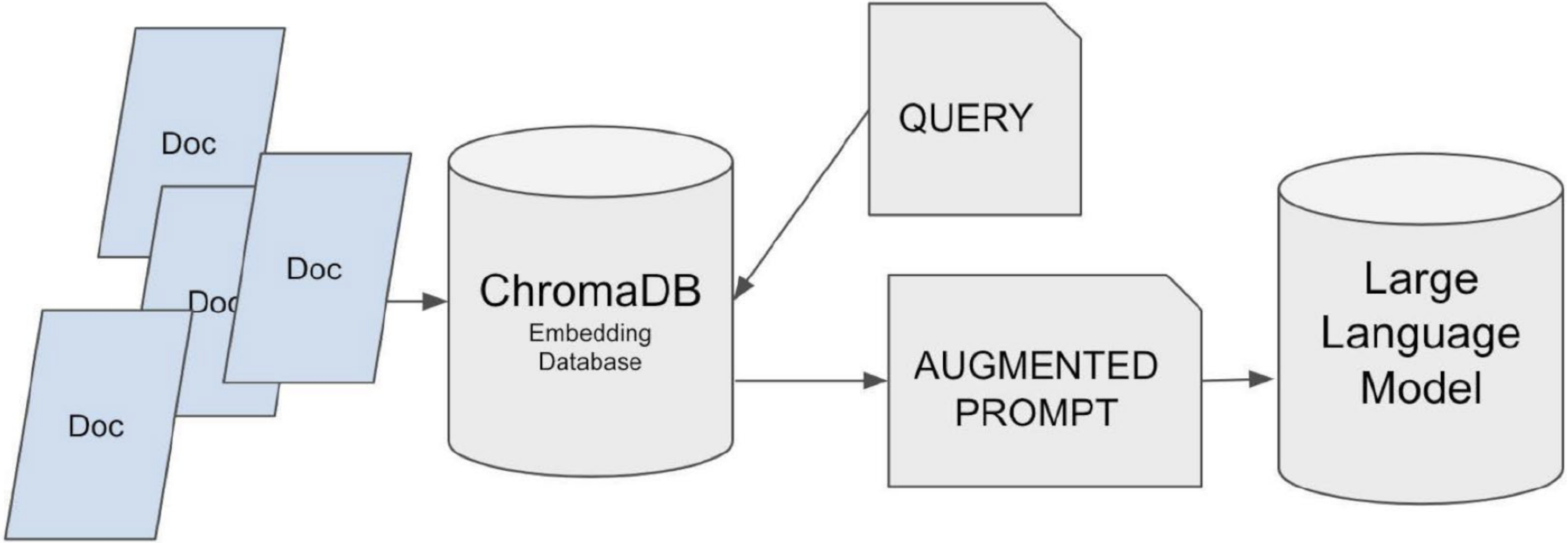
- Hub for AI and Data Science competition
- Kaggle Sections
  - Notebooks
  - Datasets
  - Models

RAG and Vector Databases

- **RAG:** exerting influence on the model's response through the integration of our own data.
- RAG Components:
  - **Knowledge Storage:** The first step is store the information that the model can draw upon. This knowledge base can include structured data (like databases), unstructured data (like text documents), or even code. We are going to use a vector database to store this information.
  - **Question Reception:** The RAG system receives the user's question or request. This could be in natural language, just like a question you'd ask a person.
  - **Information Retrieval:** This is where the "Retrieval" in RAG comes into play. The system selects relevant information to the user's question from the stored data. In a vector database, this involves vector comparison to retrieve related information from the database.

- ## RAG Components
  - **Prompt Construction:** A prompt is constructed using the relevant information and the user's question. This prompt is designed to effectively communicate the context of the user's query to the language model. In this way, not only is the model provided with information to construct its response, but hallucinations are also reduced since the model doesn't have to invent information it doesn't know.
  - **Calling the Model:** The prompt is passed to the language model. The model uses the context provided in the prompt to generate a response.
  - **Response from the Model:** The model's response is presented to the user. If the RAG system has been constructed correctly, this response should be relevant and grounded in the retrieved information from the knowledge base.

RAG and Vector Databases



ChromaDB

- Open Source
- Easy to use

ChromaDB

Table 2-1. Main features of vector databases

	ChromaDB	Faiss	Weaviate	Pinecone
Open source	Y	Y	Y	N
License	Apache 2.0	MIT	BSD-3-Clause	
Deployment	Local / Cloud	Local	Local / Cloud	Cloud
Metadata	Y	N	Y	Y
Distance metrics	Cosine / Euclidean / Dot Product	L2 / Cosine / IP / L1 / Linf	Cosine	Cosine / Euclidean / Dot Product
Algorithms	HNSW	IVF / HNSW / IMI / PQ	HNSW	Propietary

Code: LLM RAG with ChromaDB